

Lantis Red L. Fernandez

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github.com/lanseudesu

Skills

Languages: Java, Python, JavaScript, C++, C, SQL, HTML/CSS, x86 Assembly

Frameworks & Libraries: ReactJS, NextJS, NodeJS, ExpressJS, Django, Flutter

Technologies: Spring Boot, MySQL, PostgreSQL, Git, Figma, Android Studio

Experience

AltPayNet Corp.

2025

Development Security and Operations (DevSecOps) Intern

- Maintained and fixed company websites, improving functionality across web applications.
- Studied backend systems and development workflows to understand operational processes.
- Collaborated with developers to support secure coding and deployment practices.

Technical Projects

Enhanced ORB Algorithm for Retinal Image Feature Extraction

2025 - 2026

Undergraduate Thesis — Computer Vision Research

- Enhanced the ORB algorithm for retinal image analysis using adaptive thresholding, descriptor refinement, and redundancy suppression.
- Improved feature matching accuracy and keypoint distribution using Python, OpenCV, and NumPy.

Ludus Compiler – Personalized Programming Language [↗](#)

2024

Full-Stack Developer

- Developed a custom programming language with lexical analysis, parsing, semantic analysis, AST generation, and interpreter execution.
- Built runtime input handling and compiler tooling using Python, Eel, JavaScript, HTML, and CSS.

MAFA - Inventi [↗](#)

2025

Full-Stack Developer

- Developed a property management system using React, ExpressJS, and Supabase for the 2025 Inventi Hackathon.
- Implemented features for uploading property floor plans and tracking maintenance issues through an interactive map interface.

Kinetiq – Hospital Manufacturing Company Website [↗](#)

2025

Backend Developer

- Built RESTful backend services using Django and PostgreSQL and integrated frontend components with backend APIs.
- Deployed serverless infrastructure using AWS and Zappa.

Google Dinosaur Game (x86 Assembly) [↗](#)

2024

Lead Developer

- Recreated the Chrome Dinosaur Game in x86 Assembly using TASM and DOSBox.
- Implemented low-level game mechanics, collision handling, and rendering logic.

Math and Match Mobile Game [↗](#)

2024

Lead Developer

- Led the development of a Flutter-based educational memory card game combining mathematical problems with a matching-card gameplay mechanic.
- Developed the game's full-stack functionality, including the frontend, backend integration, and player data management.

Education

Pamantasan ng Lungsod ng Maynila — BS in Computer Science

Sept 2022 – Present

Expected Graduation: Aug. 2026

Extracurricular

- **Computer Science Society Creatives Committee Member**
Nov 2022 – May 2023
- **Google Developer Student Club (GDSC) Web Development Volunteer**
Oct 2024 – October 2025

Certifications

Network Support and Security Course – Cisco (2025)

Publications

Chaves, S. D., Fernandez, L. L., Marcelo, E., Mata, K. (2026). *An Enhancement of Oriented Fast and Rotated Brief (ORB) Algorithm for Feature Extraction in Retinal Images. International Journal for Multidisciplinary Research, 8(1).* DOI: 10.36948/ijfmr.2026.v08i01.70140 [↗](#)

- Undergraduate thesis research focused on enhancing the ORB feature extraction algorithm for retinal images through adaptive thresholding and improved feature descriptors.

References

Bren Alfred España — *Business Solutions Manager, JTI*
Professor, Pamantasan ng Lungsod ng Maynila

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